

SETS

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Empty sets – An interesting analogy for understanding the difference between $\emptyset$ and $\{\emptyset\}$ is to think of sets as **boxes**. Think of $\emptyset$ as an **empty box**—there is nothing inside it. Now think of $\{\emptyset\}$ as a *box containing one empty box*. The outer box is not empty because it contains the inner empty box.

Cartesian Product – The inequality $(A \times B) \times C \neq A \times B \times C$ comes down to a difference in the **structure of their elements** (the nesting of the parentheses):

- $(A \times B) \times C$ **contains ordered pairs:** Its elements look like $((a, b), c)$, where the first component is itself an ordered pair.
- $A \times B \times C$ **contains ordered triples:** Its elements look like (a, b, c) , which are flat 3-tuples.

Because a nested pair like $((a, b), c)$ is formally distinct from a 3-tuple like (a, b, c) , the two sets consist of entirely different mathematical objects and are strictly unequal. *While strictly unequal, they are naturally isomorphic or in a 1-to-1 correspondence, which is why they are often treated interchangeably in informal contexts.*

Power set – If A is a set, the **power set** of A is another set, denoted as $\mathcal{P}(A)$ and defined to be the set of all subsets of A . In symbols, $\mathcal{P} = \{X : X \subseteq A\}$. If A is a finite set, then $|\mathcal{P}(A)| = 2^{|A|}$.